20/8/26

EXPERIMENT-7

FLOW CONTROL USING SLIDING WINDOW PROTOCOL

AIM:

To write a program to implement flow control at the data link using the sliding window protocol & to simulate the flow of frames from one node to another.

ALGORITHM:

1) Start the program.

2) Read the total number of frames to be sent (N), the sender's window size (W), & typically a frame number to simulate as lost during transmission.

3) Initialize base = 0 & nextseq = 0

4) Repeat while base < N:

   a) Determine the current sender window as the range [base, base+W-1]

   b) Send all the frames with sequence numbers from nextseq upto (base+W-1).

   c) If the lost frame lies within the frames just sent:

      (i) The receiver discards the lost frame & any subsequent out of order frames.

5) Repeat step 4 until all frames have been transmitted & acknowledged.

6) Stop the program.

PROGRAM:

```
def simulate(total_frames, window_size, total_frames-1):
    print("\n")
    print(f"Sender Window: [Frame {base}] - Frame {Window_end}]")
    while next_seq < base + window_size & next_seq < total_frames:
        print(f"Node A --> sending frame {next_seq}")
        if next_seq == lost_frame:
            print(f"(Frame {next_seq}) LOST in transit)")
        next_seq += 1
    if lost_frame is not None & base <= lost_frame < next_seq:
        print(f"Node B --> Received frame {f}, sending ACK {f}")
        print(f"Node A -> ACK {next_seq} received")
        base = next_seq
        print("\n")
def main():
    print("Sliding window protocol - flow control & simulation (Go-Back-N)\n")
    total_frames = int(input())
    window_size = int(input())
    lost_frame = int(input())
    if lost_frame == -1:
        lost_frame = None
    simulate(total_frames, window_size, lost_frames)
```

OUTPUT:

Sliding Window protocol = Flow control simulation (Go-

Enter total number of frames to send : 8
Enter sender's window size = 4
Enter frame number to simulate as LOST
(0-7, enter -1 for no Loss): 5

Sender Window: [Frame 0 - Frame 3]
Node A → sending Frame 0
Node A → sending Frame 1
Node A → sending Frame 2
Node A → sending Frame 3
Node B → Received Frame 0, sending ACK 0
Node B → Received Frame 1, sending ACK 1
Node B → Received Frame 2, sending ACK 2
Node A → ACK 3 received, sliding window forwarded.

Sender Window: [Frame 4 - Frame 7]
Node A → sending Frame 5
Node A → sending Frame 5
Node A → sending Frame 6
Node A → sending Frame 7
Node B → Frame 5 missing
Node A → Timeout! Resending window starting from Frame 5.

Sender Window: [Frame 4 - Frame 7]
Node A → sending Frame 5
Node A → sending Frame 6
Node A → sending Frame 7
Node B → Received Frame 4, sending ACK 4
Node B → Received Frame 5, sending ACK 5
Node A → ACK 7 received. Sliding window forward.
All 8 frames successfully transmitted from Node A to Node B.

```
if __name__=="__main__":
    main()
```

RESULT:

The program to implement flow control at the data link layer using the sliding window protocol was successfully written in python, executed & verified.